

Local Rules Can Require Nonlocal Coordinates and Infinite-State Realizations

Limits of exact reduction in Kappa/BNGL

Research note

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Abstract

Finite Kappa and BioNetGen Language (BNGL) programs can generate countably infinite continuous-time Markov chains even though every rule is local and the program text is finite. We ask two exact-reduction questions that arise naturally in rule-based modeling: whether a designated assay process has any finite hidden-CTMC realization from one experimental seed, and whether finitely many ordinary local site-graph observables can form an assay-sufficient Markov state uniformly over reachable restart states.

The closest classical comparison is semantic finiteness of pushdown processes: for ordinary pushdown automata, deciding whether a configuration is bisimilar to some unspecified finite-state process is decidable. In contrast, ordinary Kappa has free-site tests that implement zero tests and permit two-counter-machine encodings. We show that both exact-reduction questions above are undecidable for finite Kappa programs, even under promises designed to exclude the trivial explanation that one branch merely has finite reachability.

The constructive core is a processive scanner. No finite family of finite Kappa/BNGL patterns determines its exact future assay law over all reachable restarts, although one nonlocal integer—distance remaining—does. This formalizes a gap between local molecular syntax and exact predictive state. The scanner’s designated construction seed nevertheless has a three-state exact realization, separating reusable reduction from seed-specific realization. Conversely, a unique-tip local site-graph program realizes the classical M/M/1 busy period; its non-rational completion transform rules out every finite-dimensional time-homogeneous linear realization from one finite seed. The queueing obstruction itself is classical; its role is to show what ordinary biochemical site-graph syntax can express.

Bounded reachable complex size remains a broad positive certificate for finite-dimensional exact local coordinates. Thus finite rule syntax, finite coordinate dimension, finite local expressibility, and finite hidden-state cardinality are distinct notions, with a decidable/certifiable positive frontier but no sound-and-complete decision procedure in general.

1 Questions, decision problems, and main results

The motivating question is experimental: a rule-based molecular model may contain far more microscopic information than an assay observes. When can the hidden dynamics be represented *exactly* by a finite state description that still predicts the assay?

We separate two algorithmic questions from the outset.

SEED-FINITE-HIDDEN-CTMC. Given a finite ordinary-Kappa program, a finite deterministic seed, and a finite local assay, does the assay process from that seed admit any exact finite-state time-homogeneous hidden CTMC realization?

FINITE-LOCAL-PATTERN-STATE. Given the same type of rule program and assay, does there exist a finite family of finite local site-graph patterns whose values form an assay-sufficient Markov state, with one generator depending only on those values, for every reachable deterministic restart state?

The first is a realization question about one experimental preparation. The second is a reusable static-reduction question: the same coordinate must remain exact after changes of copy number, perturbation restarts, or simulation from an intermediate state. These decision problems are restated formally in [section 6](#), where the reductions are given.

Why undecidability is not automatic. There is a close classical analogue for pushdown processes. Jančar proved that, for an ordinary pushdown-automaton configuration, it is decidable whether the induced infinite process is bisimilar to *some unspecified finite-state process*. Göller and Parys subsequently proved that whenever a pushdown system is bisimulation-finite, it has an equivalent finite system of elementarily bounded size, yielding an elementary decision procedure and improving the previous Ackermann upper bound [12, 13]. Our results therefore do not amount to the generic statement that infinite-state systems are hard. They locate a different boundary: Kappa’s ordinary free-site match is a genuine zero test, sufficient to encode deterministic two-counter machines. This is not a strict generalization of Jančar’s theorem: his problem is nonprobabilistic bisimulation/semantic finiteness, whereas our seed problem is realization-flavored and our reusable problem is tied to local rule-language observables. Probabilistic pushdown work supplies another nearby comparison: for unary pPDA, language equivalence—equivalently equality of the accepted-word-length distributions for two *given* systems—is in PSPACE and is at least as hard as polynomial identity testing [11]; that is an equivalence problem, not an existence-of-finite-realization problem.

Why the reductions are more than a bare Rice-style argument. At a high level, the undecidability proofs have the familiar Rice-style architecture: once a rule language can simulate a two-counter machine and can express assay behaviors on opposite sides of a semantic property, an unconstrained halting reduction is unsurprising. We do *not* claim that generic architecture as new, nor do we invoke Rice’s theorem literally. The work in [section 6](#) is in satisfying substantially stronger promises simultaneously. In both branches the assay-relevant reachable behavior has a countably infinite coarsest observation-compatible strong quotient; the finite-local-pattern reduction additionally has a uniform global exit-rate bound; and the NO branch of the seed-realization reduction already fails finite-dimensional realization at the level of a one-time marginal. A bare semantic-property reduction could use a trivially finite YES branch and would establish none of these separations. Accordingly, the substantive constructions are the two witnesses below and the constrained reductions that package them into decision problems.

Definition 1.1 (Observation-compatible strong lumping). *Let X_t be a CTMC on a reachable state space S and $h : S \rightarrow Y$ a deterministic assay. A partition $\mathcal{P}$ is observation-compatible if h is constant on every block. It is a strong lumping if the block process is Markov for every initial distribution.*

This is classical CTMC lumpability / Markovian bisimulation. The coarsest strong lumping refining a prescribed observation partition is not, in general, the minimal representation of the observation law. Two microscopic states may have identical assay laws while hidden intermediate rates prevent them from belonging to the same strong-lumping block. Minimum realization belongs to hidden-state / weighted-automaton theory, not to state partitions alone [5, 19, 22].

Definition 1.2 (Finite local pattern statistic). *A finite local pattern statistic is a map*

$$\Phi_F(x) = (N_{P_1}(x), \dots, N_{P_m}(x))$$

constructed from finitely many finite Kappa/BNGL site-graph patterns P_i , possibly followed by finitely many fixed deterministic functions of these counts. Here N_P denotes the ordinary occurrence/embedding count under the chosen rule-language convention.

This covers ordinary Kappa pattern observables and BNGL Molecules observables. BNGL Species observables and functional expressions are treated in [theorem 2.3](#).

Definition 1.3 (Finite hidden CTMC realization from a seed). *Fix a microscopic seed x_0 . Its assay process admits a finite hidden CTMC realization if there exist a finite state set Z , a time-homogeneous generator G , a state-dependent emission kernel, and one initial distribution α_{x_0} on Z whose emitted process has exactly the same law as $h(X_t)$ started from x_0 .*

Main results. The processive scanner in [section 2](#) proves that no finite family of ordinary finite local patterns can determine the exact future assay law over all reachable restarts, even though one nonlocal integer—distance remaining—is sufficient. This gives a precise rule-language meaning to “local rules can require nonlocal predictive state.” The same example has a three-state exact realization from its particular construction seed, so reusable reduction and seed-specific realization are genuinely distinct.

The reversible-tip construction in [section 3](#) gives the opposite separation. A single local count—active length—is an exact Markov coordinate, but from one finite seed the binary Done assay has the classical M/M/1 busy-period law and admits no finite-dimensional time-homogeneous linear realization. Thus finite coordinate dimension does not imply finite hidden-state cardinality.

Finally, [section 6](#) proves both decision problems undecidable by explicit two-counter reductions. The reductions are arranged so that in *both* YES and NO branches the coarsest observation-compatible strong quotient is already countably infinite. The hardness is therefore not a disguised finite-versus-infinite reachability test. This matters for static analysis: KaSa and related Kappa analyses already compute sound abstractions and can certify non-polymerization in useful cases [3]; our theorem rules out a sound-and-complete checker for exact finite local assay state in full Kappa, while [theorem 4.1](#) identifies a broad positive certified subclass.

Novelty position. The strong-lumping, event-bisimulation, overlap-algebra, Erlang, M/M/1 busy-period, matrix-exponential, and pushdown generating-function ingredients are background. Kappa already has exact internal coarse-graining, stochastic fragments, abstract differential semantics, intrinsic information-carrier analyses, counter abstractions, and rule-algebraic observable dynamics [15, 7, 18, 16, 4, 28, 1, 2]. Queueing, stochastic-Petri-net, recursive-Markov-chain, and probabilistic-pushdown theories likewise contain the classical infinite-state and algebraic termination phenomena used for calibration [27, 26, 24, 10, 11]. Undecidability for Kappa-presented CTMCs is also not new in itself: Danos and Oury reduce Post correspondence to an equilibrium/dissipativity question for Kappa graph rewriting [6].

The narrow claims defended here are therefore the two rule-language witnesses and the constrained reductions built from them: (i) a finite local site-graph program can require a nonlocal coordinate because no finite family of ordinary finite patterns determines all future assay laws of its reachable restart states; (ii) ordinary local site-graph syntax can realize a seed-level assay with no finite-dimensional exact linear realization; and (iii) the corresponding exact-reduction existence problems are undecidable under promises that rule out the cheap finite-versus-infinite explanation. The M/M/1 calculation used for (ii) is explicitly classical probability embedded in biochemical rule syntax, not a new queueing theorem. Formal CTMC background and its proofs are deferred to the appendices so that the rule-language results appear first.

2 A one-way processive scanner: local patterns do not determine position

We use the standard stochastic rule semantics summarized in Appendix A; in particular, q_x denotes the total CTMC exit rate from microscopic mixture x . Consider a finite local Kappa program with a homogeneous linear A -polymer, a unique scanner S , a unique writer W , and a controller C . In build mode the writer extends one marked end at rate λ_b ; Freeze switches to scan mode at rate λ_s . In scan mode the scanner moves one monomer toward the marked terminal end at rate κ , and at the terminal monomer a final local rule sets a Done bit, also at rate κ . The scanner and writer are unique, so there is at most one enabled event of each named type. A finite seed contains one monomer, scanner and writer on that monomer, and controller in build mode.

This can be written with ordinary finite Kappa site graphs. The relevant constructs have direct BNGL analogues: a linear molecule/agent with left/right binding sites, a unique bound marker, internal controller states, synthesis of one new monomer by the append rule, and a local Done observable. The theorem uses only the induced local match structure, not syntax peculiarities of one implementation.

For a scan-mode state, let $r \geq 1$ be the number of rate- κ events remaining until Done, including the final Done event. Let B_r be the set of reachable scan states with this value, and B_0 the Done states. Every state in B_r has one rate- κ transition into B_{r-1} and no other assay-relevant transition.

From any $x \in B_r$, the Done hitting time is Erlang(r, κ), so

$$S_r(t) = \mathbb{P}_x\{h(X_t) = 0\} = e^{-\kappa t} \sum_{j=0}^{r-1} \frac{(\kappa t)^j}{j!}. \quad (1)$$

2.1 No finite local pattern vocabulary is a reusable exact state

Lemma 2.1 (Finite local patterns cannot locate a deep interior scanner). *Let F be any finite family of finite ordinary Kappa site-graph patterns, and let A be any fixed ambient mixture disjoint from a homogeneous scanner chain. There exists $R_F < \infty$ such that, on every sufficiently long scanner chain adjoined to A , any two scanner positions whose graph distance from both chain ends exceeds R_F have identical F -count vectors. The same R_F works independently of the size or internal structure of the fixed ambient mixture.*

Proof. For a connected pattern P , an embedding lies wholly inside one connected component. Embeddings into the fixed ambient mixture A therefore contribute the same additive constant in both compared states. Inside the scanner chain, every embedding occupies a subgraph of diameter at most $|P| - 1$. If an embedding contains the scanner, it inspects only a bounded rooted neighborhood; sufficiently far from both ends this neighborhood is translation-invariant. If it does not contain the scanner, moving the unique scanner defect between two deep interior sites only translates that local defect in an otherwise homogeneous path, leaving total connected-motif counts unchanged. Boundary markers can matter only when the pattern reaches them.

For disconnected patterns, use the standard graph-overlap product identity: products of connected-pattern counts equal the disjoint-union count plus finitely many consistent overlap/gluing counts. Induction expresses every finite pattern count as a polynomial in finitely many connected-pattern counts of bounded size, as in rule-algebraic observable calculations [1, 2]. Because all relevant connected counts agree in the two mixtures, so do all disconnected-pattern counts. Taking R_F larger than all required diameters proves the claim. $\square$

Theorem 2.2 (No finite local pattern family determines the future assay law). *For every finite family F of finite ordinary Kappa site-graph patterns, there exist two reachable deterministic scanner states x, x' such that*

$$\Phi_F(x) = \Phi_F(x'), \quad h(x) = h(x'),$$

but

$$\mathcal{L}_x(h(X.)) \neq \mathcal{L}_{x'}(h(X.)).$$

Consequently no finite local pattern statistic is sufficient for the exact future assay law over all reachable restarts. A fortiori, no such statistic can be an observation-compatible time-homogeneous Markov state with one generator depending only on its value.

Proof. Choose one sufficiently long chain and two different deep interior scanner positions on it; take the same fixed ambient mixture in both states. By [theorem 2.1](#) they have identical Φ_F values. Their remaining distances $r \neq r'$ give different future assay survival laws $S_r \neq S_{r'}$ by [equation \(1\)](#). Thus equality of the finite local statistic does not determine even the one-time future assay marginals, hence cannot determine the full future assay law. The Markov-state conclusion follows immediately. $\square$

Remark 2.3 (BNGL observable vocabulary). *The argument is not restricted to Kappa embedding counts or BNGL Molecules. On this single connected scanner complex, any fixed finite collection of BNGL Species observables built from finite local patterns is likewise insensitive to sufficiently deep interior position: whether the one complex contains a given finite motif is determined by a bounded neighborhood. Any fixed functional expression of finitely many such local observables remains equal when its inputs are equal. What escapes the theorem is an explicitly nonlocal counter such as “distance remaining.”*

Remark 2.4 (Why Kappa counters are the natural escape hatch). *Theorem 2.2 formalizes an intuition that motivated explicit counter extensions of rule languages. Ordinary finite site-graph patterns can inspect only bounded molecular neighborhoods; they cannot encode an unbounded processive distance. Kappa counters add an integer-valued coordinate directly to the language, precisely the kind of nonlocal state that closes the scanner. The counter extension is therefore not merely syntactic convenience here: it supplies a representation class that ordinary local patterns provably lack [4].*

The coarsest observation-compatible strong quotient of the scanner is nevertheless easy to describe: build states are indexed by current chain length, scan states by remaining distance, plus Done. It is countably infinite. Appendix B proves that the standard observation-seeded refinement reaches this quotient only at the ω limit; no finite refinement depth suffices.

Proposition 2.5 (Uniform nonexplosion). *The scanner CTMC satisfies*

$$q_x \leq \lambda_b + \lambda_s + \kappa$$

for every reachable state.

Proof. In build mode there is one writer match and one Freeze match; in scan mode the unique scanner enables exactly one move or Done event. A CTMC with uniformly bounded exit rate is nonexplosive. $\square$

2.2 All-reachable-state linear rank

Lemma 2.6 (Linear independence). *The family $\{S_r : r \geq 1\}$ is linearly independent over $\mathbb{R}$.*

Proof. If $\sum_{r=1}^R c_r S_r(t) = 0$, multiply by $e^{\kappa t}$. The result is a polynomial identity. If R is maximal with $c_R \neq 0$, the coefficient of t^{R-1} is $c_R \kappa^{R-1}/(R-1)!$, contradiction. $\square$

Proposition 2.7 (No finite hidden model works for every reachable restart). *No finite-state hidden CTMC can reproduce the scanner assay laws simultaneously for every reachable deterministic initial state, even allowing an arbitrary hidden initialization distribution for each microscopic restart and arbitrary stochastic emissions. More strongly, no finite-dimensional time-homogeneous linear system can contain all one-time marginal functions S_r .*

Proof. For a K -dimensional linear realization, every output has the form $u_r^\top e^{Gt}b$ and therefore belongs to the span of the K coordinate functions of $e^{Gt}b$. This span has dimension at most K , contradicting [theorem 2.6](#). A finite hidden CTMC is a special case. $\square$

The linear independence is classical phase-type mathematics: Erlang order grows with r . The rule-based content is that a single finite, local, uniformly bounded Kappa program contains all these orders among its reachable restart states.

2.3 The construction seed is different: an exact three-state collapse

The all-reachable quantifier above is substantive, not cosmetic. From the scanner's actual seed, let T be the first Done time. During build, Append and Freeze race at rates λ_b and λ_s ; after Freeze at chain length n , scanning requires n independent rate- κ events.

Proposition 2.8 (Seed-specific three-state realization). *From the construction seed,*

$$\mathbb{E}[e^{-sT}] = \frac{\lambda_s \kappa}{s^2 + (\lambda_b + \lambda_s + \kappa)s + \lambda_s \kappa}. \quad (2)$$

Hence the binary assay process from that seed has an exact three-state hidden CTMC realization (two transient phases emitting 0, one absorbing Done state emitting 1).

Proof. Let $F_n(s)$ be the completion-time transform in build mode with chain length n , and set

$$u = \frac{\lambda_b}{\lambda_b + \lambda_s + s}, \quad v = \frac{\lambda_s}{\lambda_b + \lambda_s + s}, \quad w = \frac{\kappa}{\kappa + s}.$$

Conditioning on the first build-mode event gives

$$F_n = uF_{n+1} + vw^n.$$

The recurrence has general solution $F_n = vw^n/(1-uw) + Cu^{-n}$. Since $0 < u < 1$ and every Laplace transform satisfies $0 \leq F_n \leq 1$, boundedness as $n \rightarrow \infty$ forces $C = 0$. Hence the unique bounded solution is $F_n = vw^n/(1-uw)$, which at $n = 1$ simplifies to [equation \(2\)](#). The quadratic with sum $\lambda_b + \lambda_s + \kappa$ and product $\lambda_s \kappa$ has positive roots a, b : its discriminant satisfies

$$(\lambda_b + \lambda_s + \kappa)^2 - 4\lambda_s \kappa \geq (\lambda_s - \kappa)^2,$$

with strict inequality when $\lambda_b > 0$, and the sum and product are positive. Thus

$$a + b = \lambda_b + \lambda_s + \kappa, \quad ab = \lambda_s \kappa.$$

Then [equation \(2\)](#) equals $a/(s+a) b/(s+b)$, the transform of the sum of two independent exponentials. (If the roots coalesce in a degenerate limiting parameter choice, the law is Erlang-2 and remains two-phase.) Because the assay is exactly 0 before T and 1 after T , the corresponding two-phase absorbing CTMC reproduces its entire path law. $\square$

Thus the scanner proves a strong reusable-state obstruction, but *not* a seed-specific finite-realization obstruction. The next witness supplies the missing statement.

3 A local site-graph realization of the M/M/1 busy period

Consider a second finite local Kappa program encoding a rooted active polymer with a unique marked tip. Its active length is $R_t \in \mathbb{N}_0$. A *Push* rule appends one active monomer at the marked tip and moves the tip marker, at rate $\lambda > 0$. A *Pop* rule removes (or detaches and marks inert) the terminal active monomer and moves the tip marker to its predecessor, at rate $\mu > \lambda$. When the last active monomer is popped, a local root-tip rule sets a Done bit and no further active rule fires. Start from one active monomer.

This requires only finitely many finite site-graph rules. To avoid deletion side effects, popped monomers may simply be detached and changed to an internal inert state; no other rule mentions that state. Schematically, with a root R , active monomer A , internal flag $t \in \{0, 1\}$ for the unique tip, and an inert flag i , the three rule forms are

$$\begin{aligned} \text{Push:} \quad & A(t=1, r) \longrightarrow A(t=0, r!1), A(t=1, l!1), \\ \text{Pop}_{\geq 2}: \quad & A(t=0, r!1), A(t=1, l!1) \longrightarrow A(t=1, r), A(i=1), \\ \text{Pop}_1: \quad & R(x!1), A(t=1, l!1) \longrightarrow R(\text{done}), A(i=1), \end{aligned}$$

with rates λ, μ, μ , respectively; omitted sites are unchanged. The display is semantic shorthand rather than executable syntax, but each line is an ordinary finite Kappa rewrite. BNGL uses the same ingredients—molecule components, internal states, explicit bonds, and product-only molecule creation—so the induced birth-death length process has a direct BNGL/NFsim counterpart.

The unique tip guarantees exactly one Push and one Pop match whenever $R_t \geq 1$. Hence, before Done,

$$r \longrightarrow r + 1 \text{ at rate } \lambda, \quad r \longrightarrow r - 1 \text{ at rate } \mu, \quad (3)$$

with 0 absorbing and assay $h = \mathbf{1}\{R_t = 0\}$. This is exactly the queue-length chain during the classical stable M/M/1 busy period, started by one customer [27]. The site-graph point is not the birth-death calculation itself: the unique tip marker makes both increment and decrement hazards copy-number independent using ordinary local rewrite rules. In a mass-action molecular description without such a distinguished reactive tip, a nominal “decrement” reaction would ordinarily acquire a multiplicity-dependent hazard.

Proposition 3.1 (Uniform nonexplosion). *The reversible-tip program has global exit-rate bound $q_x \leq \lambda + \mu$.*

Proof. The unique tip enables exactly one Push and exactly one Pop event at each positive length; at Done neither is enabled. $\square$

Lemma 3.2 (Completion-time transform). *Let $T = \inf\{t : R_t = 0\}$ with $R_0 = 1$. For $s \geq 0$,*

$$\rho(s) := \mathbb{E}[e^{-sT}] = \frac{s + \lambda + \mu - \sqrt{(s + \lambda + \mu)^2 - 4\lambda\mu}}{2\lambda}. \quad (4)$$

The function ρ is not rational.

Proof. Let $f_r(s) = \mathbb{E}_r[e^{-sT}]$. First-step analysis gives, for $r \geq 1$,

$$(s + \lambda + \mu)f_r = \lambda f_{r+1} + \mu f_{r-1}, \quad f_0 = 1.$$

Translation invariance before hitting the next lower level, plus the strong Markov property, gives $f_r = \rho^r$; equivalently substitute this ansatz into the recurrence. The quadratic is

$$\lambda \rho^2 - (s + \lambda + \mu)\rho + \mu = 0.$$

Because $\mu > \lambda$, the bounded root satisfying $\rho(0) = 1$ is [equation \(4\)](#).

Its discriminant

$$\Delta(s) = (s + \lambda + \mu)^2 - 4\lambda\mu$$

has two distinct simple zeros $-(\sqrt{\mu} - \sqrt{\lambda})^2$ and $-(\sqrt{\mu} + \sqrt{\lambda})^2$. A square of a rational function has zeros and poles of even order, so $\sqrt{\Delta(s)}$ is not rational; neither is $\rho(s)$. $\square$

Theorem 3.3 (Single-seed no finite-dimensional realization). *The reversible-tip Done assay from its single finite seed admits no finite-dimensional time-homogeneous linear realization. Consequently it admits no finite-state time-homogeneous hidden CTMC realization, even if the hidden model is allowed arbitrary stochastic emissions. This failure is already visible in the one-time marginal $S(t) = \mathbb{P}\{h(X_t) = 0\}$.*

Proof. Because $S(t) = \mathbb{P}(T > t)$,

$$\widehat{S}(s) = \int_0^\infty e^{-st} S(t) dt = \frac{1 - \rho(s)}{s}. \quad (5)$$

By [theorem 3.2](#), this is non-rational: if $(1 - \rho(s))/s$ were rational, then $\rho(s) = 1 - s\widehat{S}(s)$ would be rational as well.

If a finite-dimensional time-homogeneous linear realization existed, then for some finite matrix G and vectors u, b ,

$$S(t) = u^\top e^{Gt} b.$$

No positivity, stochasticity, or generator sign constraints are being used here. For sufficiently large real s ,

$$\widehat{S}(s) = u^\top (sI - G)^{-1} b,$$

a rational function of s , contradiction. A finite hidden CTMC marginal has exactly this matrix-exponential form, so it is also impossible. $\square$

Remark 3.4 (Classical queueing content). *The transform in [theorem 3.2](#) is the classical Takács M/M/1 busy-period transform [27]. Likewise, a finite-dimensional matrix-exponential law has a rational Laplace transform [26]. We therefore do not claim the non-rationality obstruction as new probability. Its role is to show, in the same local site-graph syntax used for biochemical rule models, that a finite rule set and finite seed need not imply a finite-state exact assay realization.*

Corollary 3.5 (Exact half-integer tail and unbounded approximation order). *The completion-time density of the reversible-tip program has asymptotic form*

$$f_T(t) \sim C t^{-3/2} \exp \left[-(\sqrt{\mu} - \sqrt{\lambda})^2 t \right] \quad (t \rightarrow \infty)$$

for a constant $C > 0$. *This is an exact model-class signature, not an empirical falsification criterion: phase-type distributions are dense among distributions on $[0, \infty)$, so finite-state models can approximate the law arbitrarily closely in weak topology. However, any sequence of phase-type approximations converging weakly to this busy-period law must have unbounded phase order.*

Proof. For the M/M/1 busy period,

$$f_T(t) = \sqrt{\frac{\mu}{\lambda}} \frac{1}{t} e^{-(\lambda+\mu)t} I_1(2t\sqrt{\lambda\mu}),$$

where I_1 is the modified Bessel function. The standard asymptotic $I_1(z) \sim e^z/\sqrt{2\pi z}$ gives the displayed form. The nonexistence of an exact finite-dimensional linear realization already follows from [theorem 3.3](#); the tail is an interpretable manifestation of the same obstruction rather than an independent proof.

For the approximation statement, denseness of the phase-type class is classical [9, 26]. Suppose, toward contradiction, that phase-type laws H_n of uniformly bounded order K converge weakly to the busy-period law. Their Laplace transforms $r_n(s)$ converge pointwise for every $s \geq 0$ to $\rho(s)$, because e^{-st} is bounded and continuous. Each $r_n = p_n/q_n$ is rational with numerator and denominator degrees bounded by K . Normalize the joint coefficient vector of (p_n, q_n) to Euclidean norm one. By compactness of the unit sphere in a fixed finite-dimensional coefficient space, pass to a subsequence with coefficientwise limits (p, q) , not both zero. From $q_n(s)r_n(s) - p_n(s) = 0$ and pointwise convergence, $q(s)\rho(s) = p(s)$ for every $s > 0$. If q were identically zero, then p would vanish on $(0, \infty)$ and hence identically, contradiction. Thus $\rho = p/q$ wherever $q \neq 0$, making ρ rational by analytic continuation, contrary to [theorem 3.2](#). Hence phase order cannot remain bounded along a convergent approximation sequence. $\square$

4 A positive boundary: bounded complexes

The negative theorems do not say that countably infinite rule-based CTMCs lack finite-dimensional exact coordinates.

Theorem 4.1 (Bounded-complex certificate). *Suppose there is $B < \infty$ such that every connected molecular complex in every reachable mixture has at most B agents. Then finitely many fully specified connected species-count coordinates form an exact Markov coordinate system for the complete microscopic process.*

Proof. A finite signature and size bound imply that only finitely many connected fully specified species graphs $C_1, \dots, C_M$ can occur up to isomorphism. A mixture is uniquely determined by its multiplicity vector

$$\Phi(x) = (n_{C_1}(x), \dots, n_{C_M}(x)) \in \mathbb{N}^M.$$

Thus Φ is injective and therefore exact. Its state space may be infinite because copy numbers are unbounded, but its coordinate dimension is finite. $\square$

The reversible-tip process has a one-dimensional exact local coordinate—the active monomer count—even though no finite-state realization exists. The processive scanner has a one-dimensional exact *nonlocal* coordinate—distance remaining—but no finite local pattern family computes it. Thus finite dimension, finite state cardinality, and finite local syntax are genuinely different axes.

5 Ordinary Kappa can simulate a deterministic two-counter machine

The undecidability reductions below do not rely on the claim that the Kappa counter paper itself proves two-counter-machine undecidability. Instead we give the encoding directly. The paper by

Boutillier, Cristescu, and Feret establishes Kappa counter semantics and compilation machinery and is relevant precedent for representing counters in the language [4]; undecidability comes from the classical two-counter machine [21, 14].

Take a deterministic two-counter machine with finite control states and two nonnegative counters, started from the designated all-zero configuration. We use the CM2-style instruction family for which halting is undecidable [14], written here in the equivalent explicit-control form:

- (i) $\text{INC}(i, q')$: increment counter i and go to q' ;
- (ii) $\text{JZDEC}(i, q_0, q_1)$: if counter i is zero go to q_0 , otherwise decrement it and go to q_1 .

There is one halt state. Dudenhefner’s CM2 syntax is a syntactic subfamily of this explicit-control notation: “next instruction” is represented by the corresponding fixed successor label. If a halting instance is supplied with nonzero initial counters, a finite prefix of INC instructions loads those values, so undecidability transfers to the designated all-zero initial configuration used below.

Lemma 5.1 (Finite ordinary-Kappa encoding). *For every deterministic two-counter machine M there is a finite ordinary-Kappa site-graph program and finite seed whose reachable non-halted configurations project one-to-one onto the unique run of M . Each simulated machine configuration enables exactly one instruction rule variant. All instruction variants may be assigned the same constant rate $\gamma > 0$.*

Proof. Represent counter i by a rooted oriented chain. Use a unique root agent $R_i(s)$ and token agents $U_i(p, n)$. Zero is represented by the root site s being free. A positive value m is represented by

$$R_i - U_{i,1} - U_{i,2} - \cdots - U_{i,m},$$

with orientation encoded by the p/n sites. A unique controller agent carries the finite program-counter internal state.

For INC, use two local rule variants. If the root is free, create one token and bind it to the root. If the root is bound to the first token, insert one fresh token between root and first token. Both update the controller to q' . For JZDEC, the zero branch matches the *free* root site and updates the controller to q_0 . The nonzero branch matches the root and first token, detaches that token (marking it inert if one wishes to avoid deletion), reconnects the root to the second token when one exists, and updates to q_1 . One-token and multi-token cases require two finite local variants.

The free-site match is exactly the zero test: zero and nonzero patterns are mutually exclusive. Likewise, one-token and multi-token pop patterns are mutually exclusive. Hence every encoded machine configuration enables exactly one instruction rule variant, and the resulting graph represents exactly the successor machine configuration. There are finitely many rule variants because the machine has finitely many instructions. The invariant is immediate by induction on simulated steps: exactly one controller exists, each counter has exactly one root and an oriented token chain of the encoded length, zero is equivalent to a free root site, and precisely the rule variant representing the machine’s unique next instruction is enabled. Assigning all variants rate γ changes only event timing, not the unique discrete run. No probabilistic simulation claim is needed: reachability follows the deterministic machine run exactly. $\square$

The free-site zero test plays the role of an inhibitor condition. This is the computational feature absent from plain vector-addition/Petri-net dynamics, whose boundedness admits Karp–Miller style decision procedures [20]; we use it explicitly rather than hiding the hardness inside a citation.

6 Undecidability of exact realization

The contrast with pushdown semantic finiteness explains why these theorems are not automatic. For ordinary pushdown processes, deciding whether a configuration is bisimilar to some unspecified finite-state process is decidable [12, 13]. Kappa, however, admits a local free-site test that acts as a zero test and raises the expressive power to deterministic two-counter machines. The reductions below show that the analogous exact-reduction existence questions become undecidable in this richer rule language.

Kappa is already known to support other undecidable semantic questions: Danos and Oury encoded Post correspondence into a Kappa CTMC and proved undecidability of an equilibrium/dissipativity property [6]. The point here is therefore property-specific rather than computational universality. We decide neither general reachability nor boundedness; instead we ask whether a designated assay possesses an exact finite realization of one of two prescribed kinds, and we force both reduction branches to remain assay-relevantly infinite.

Why ask for such a decision procedure at all? Kappa static analysis already computes sound abstractions of reachability, influence, fragments, and polymerization structure, and can certify that some agent classes do not polymerize [3]. It is therefore natural to ask whether a sound-and-complete static checker could decide that finitely many local patterns form an exact assay-sufficient state. Theorem 6.5 rules out such a checker in general. This does *not* rule out useful sound incomplete criteria.

There is also a positive decidable frontier. On any subclass for which a sound static analysis certifies a uniform bound on reachable complex size, theorem 4.1 constructs a finite exact species-count coordinate immediately; every certified instance is therefore a YES instance of the finite-local-coordinate problem. The open algorithmic target is consequently not completeness, but stronger sound certificates—for example syntactic detection of unbounded assay-relevant distance between a processive marker and its target.

6.1 Seed-level finite hidden-state realization

Definition 6.1 (SEED-FINITE-HIDDEN-CTMC). *Input: a finite ordinary-Kappa program with rational positive constant rule rates, a finite deterministic seed, and a finite local binary assay. Question: does the assay process from that designated seed admit any exact finite-state time-homogeneous hidden CTMC realization?*

Theorem 6.2 (Seed-level finite hidden-state realization is undecidable). *SEED-FINITE-HIDDEN-CTMC is undecidable. The result remains true under the promise that both YES and NO instances have infinitely many reachable states with distinct future assay laws; for the constructions below, the coarsest observation-compatible strong quotient is countably infinite in both branches. In the NO branch, even the single one-time assay marginal admits no finite-dimensional time-homogeneous linear realization.*

Proof. Reduce from halting, from the designated all-zero configuration, of a deterministic two-counter machine with INC/JZDEC instructions. This halting problem is undecidable for the CM2 instruction family used here [14]. Encode the machine by theorem 5.1.

Seed one inert *Step* agent. Instrument every nonhalting machine-instruction rule so that, when it fires, it creates exactly one additional *Step* agent. Add one unique *Probe* agent with internal state *live/done*; every nonhalting machine instruction requires *Probe(live)*. Add the local finite rule

$$\text{Probe(live)} + \text{Step} \longrightarrow \text{Probe(done)} + \text{Step}$$

at per-match rate $\kappa > 0$. With n Step agents and a live probe, this rule has aggregate hazard κn . The assay is Probe(done).

If a machine instruction would enter the halt control state while Probe is still live, modify that finitely many rule variant(s) so that the transition leaves the live machine-controller mode (all counter-machine rules require that mode), disables the Step detector, and initializes the construction seed of the processive build/freeze scanner of [section 2.3](#); its terminal scanner event sets the same Probe(done) assay. This is a finite local syntactic modification. If the designated machine starts halted, initialize that scanner directly.

The construction is nonexplosive. Before either Probe completion or machine halt, machine instructions occur at total rate γ and the Probe transition can fire at most once, so the number of pre-halt jumps on every finite interval is bounded by a rate- γ Poisson count plus one. After halt, the process is the processive scanner, whose build and scan modes have bounded exit rates $\lambda_b + \lambda_s$ and κ , respectively.

Suppose M halts after H instructions. Before the halt transition, only the finitely many machine configurations on that unique run are relevant. Probe may absorb from any of them. Conditional on surviving to the halt transition, the underlying Kappa process enters the unbounded build/freeze scanner. By [theorem 2.8](#), the assay law from that scanner seed has an exact three-state hidden CTMC realization. Joining those three hidden states to the finite pre-halt chain gives a finite hidden CTMC exact from the designated seed.

Importantly, the halting branch is nevertheless assay-relevantly infinite: after the halt transition, arbitrarily many Append events can occur before Freeze with positive probability, and the build states of different chain lengths have different completion-time transforms $F_n(s) = vw^n/(1-uw)$. Thus infinitely many reachable microscopic states have distinct future assay laws even in the YES branch.

Now suppose M never halts. Before Probe becomes done, exactly one machine instruction is enabled at rate γ . Couple the model to the un-killed instruction process obtained by suppressing only the Probe transition; in that process the number of Step agents is

$$N_t = 1 + P_t,$$

where P_t is a Poisson process of rate γ . Conditional on this path, survival of the live probe to time t contributes

$$\exp\left(-\kappa \int_0^t N_s ds\right).$$

Therefore, using the Laplace functional of a Poisson process,

$$S(t) := \mathbb{P}\{\text{Probe is live at } t\} \tag{6}$$

$$= \mathbb{E} \exp\left(-\kappa \int_0^t (1 + P_s) ds\right) \tag{7}$$

$$= e^{-\kappa t} \exp\left(\gamma \int_0^t [e^{-\kappa(t-u)} - 1] du\right) \tag{8}$$

$$= \exp(\gamma/\kappa) \exp[-(\gamma + \kappa)t] \exp[-(\gamma/\kappa)e^{-\kappa t}]. \tag{9}$$

Put $a = \gamma/\kappa$. Expanding the final exponential gives

$$S(t) = e^a \sum_{m=0}^{\infty} \frac{(-a)^m}{m!} e^{-(\gamma + \kappa + m\kappa)t}. \tag{10}$$

Hence its Laplace transform has meromorphic continuation

$$\hat{S}(z) = e^a \sum_{m=0}^{\infty} \frac{(-a)^m}{m!} \frac{1}{z + \gamma + \kappa + m\kappa}. \quad (11)$$

The series converges locally uniformly away from the displayed poles because of the factorial denominator. At

$$z_m = -(\gamma + \kappa + m\kappa)$$

only the m th summand is singular, with nonzero residue $e^a(-a)^m/m!$. Thus $\hat{S}$ has infinitely many poles and is not rational. Any finite-dimensional time-homogeneous linear realization has rational Laplace transform $u^\top(zI - G)^{-1}b$, so no such realization exists; in particular no finite hidden CTMC can reproduce the designated-seed assay process.

The NO branch also has infinitely many reachable states with distinct future assay laws: after n machine steps and before Probe completion, the instantaneous assay-completion hazard is $\kappa(n + 1)$, which differs with n . The YES branch has infinitely many distinct future laws by the scanner argument above. Consequently, in both branches the coarsest observation-compatible strong quotient is itself countably infinite: states with different future assay laws cannot occupy one observation-compatible strong-lumping block. Therefore

$$M \text{ halts} \iff K(M) \in \text{SEED-FINITE-HIDDEN-CTMC}$$

even under the stated assay-relevant infinitude promise. Halting is undecidable, so the realization problem is undecidable. $\square$

Remark 6.3 (Why this is not a boundedness reduction). *Both branches now contain infinitely many states that the future assay law can distinguish. In the halting branch, the processive scanner has an infinite family of build lengths but its seed mixture collapses algebraically to a finite hidden realization. In the nonhalting branch, the detector marginal itself has infinitely many Laplace poles. The reduction therefore uses a genuine distinction between infinite assay-relevant state and finite seed-level realization, rather than adding an assay-blind infinite decoration. The computational hardness still enters through the counter-machine zero test; the analytic construction transfers that hardness to the exact-realization property.*

6.2 Finite local-pattern exactness in the reusable sense

Definition 6.4 (FINITE-LOCAL-PATTERN-STATE). *Input: a finite ordinary-Kappa program, finite seed, and finite local assay. Question: does there exist a finite family of finite local site-graph patterns whose value is an assay-sufficient Markov state, with one generator depending only on that value, for every reachable deterministic restart state?*

Theorem 6.5 (Finite local-pattern exactness is undecidable). *FINITE-LOCAL-PATTERN-STATE is undecidable. The result holds even under the promise that (i) in both YES and NO instances the coarsest observation-compatible strong quotient is countably infinite, and (ii) the global exit rate is uniformly bounded.*

Proof. Again encode a deterministic two-counter machine. Instrument each nonhalting instruction with one local append to a unique marked step chain. At every nonhalt control state, enable a Freeze rule at rate σ that disables the machine and activates the one-way processive scanner on the step chain. Machine instruction variants all have rate γ . The assay is a common Done bit.

If an instruction enters the halt state before Freeze, disable the machine and initialize the reversible-tip process of [section 3](#), with its absorption at active length zero setting the same Done bit. Thus in the halt branch the post-halt molecular state space is unbounded and assay-relevant, but the single local pattern count

$$N_{A_{\text{active}}}$$

is an exact Markov coordinate for that post-halt process.

By [theorem 5.1](#), every run-mode configuration enables exactly one instruction variant and one Freeze event. Scan mode enables exactly one scanner event. Reversible-tip mode enables exactly one Push and one Pop event. Therefore the total exit rate is globally bounded by

$$\max\{\gamma + \sigma, \kappa, \lambda + \mu\}.$$

If M halts after H instructions, every pre-halt counter and step chain has length at most $H + 1$, and every branch produced by Freeze before halt has scanner-chain length at most $H + 1$. Finitely many fully specified finite patterns can therefore distinguish all such pre-halt/frozen configurations. Add the finite mode/controller patterns and the one active-monomer count $N_{A_{\text{active}}}$ for the post-halt reversible tip. The lumping check is direct. Fibers in run mode and in every pre-halt frozen-scanner mode are singletons because the bounded configurations are fully specified. In tip mode, equal coordinate values can differ only by detached inert monomers, whose number affects neither any enabled rule nor any rule hazard nor the assay. The only cross-mode halt transition originates from a singleton run-mode fiber. Thus all aggregate macro-event rates depend only on the finite coordinate, and this family is an assay-sufficient Markov state for every reachable restart. The YES branch nevertheless has infinitely many distinct future assay laws because reversible-tip states of lengths $r = 1, 2, \dots$ have completion transforms $\rho(s)^r$.

If M does not halt, then for every n the deterministic run reaches its n th configuration, after which Freeze may occur. Hence scanner states on arbitrarily long step chains are reachable. Fix one such frozen machine/counter ambient mixture and choose two sufficiently deep scanner positions on its chain. By [theorem 2.2](#), any proposed finite local pattern family identifies two such states while their future Done laws differ. Therefore no finite local-pattern statistic can be assay-sufficient over all reachable restarts. The answer is YES exactly when M halts, proving undecidability under both promises. $\square$

Remark 6.6 (The two decision problems are genuinely different). *The halting branch used in [theorem 6.5](#) makes the distinction concrete. Its post-halt reversible-tip process is a YES instance for finite local-coordinate exactness because active length is a one-pattern Markov coordinate on that branch, yet by [theorem 3.3](#) the same reversible-tip seed is a NO instance for finite hidden-CTMC realization. Finite local coordinate dimension therefore does not imply finite hidden-state cardinality.*

Remark 6.7 (Why the zero test matters). *Plain Petri-net/vector-addition boundedness is decidable by Karp–Miller style constructions, whereas a free-site test acts as a true zero test. The reduction’s computational power enters exactly there. The Kappa counters paper is relevant for counter representation and compilation, not as a substitute for this undecidability proof [[4](#), [20](#)].*

7 Relation to the Kappa and reduction literature

Strong lumping and event bisimulation. Observation-seeded coarsest strong lumpings belong to classical probabilistic process theory. Event bisimulation and invariant event sigma-algebras were developed for labelled Markov processes by Danos, Desharnais, Laviolette, and Panangaden

[5]; finite and countable CTMC aggregation has a broad literature, including direct applications to combinatorial reaction networks [17, 28]. [Theorem B.1](#) should therefore be read as a concise specialization plus a sharpness witness, not as invention of the underlying bisimulation notion.

Recursive and pushdown stochastic processes. The reversible-tip witness is naturally a probabilistic one-counter / pushdown-style process. Recursive Markov chains are finite presentations of denumerable Markov chains whose reachability and termination probabilities solve monotone nonlinear systems and can be irrational [10]. For unary probabilistic pushdown automata, Forejt, Jančar, Kiefer, and Worrell show that language equivalence coincides with equality of termination-time distributions; the problem is in PSPACE and is at least as hard as polynomial identity testing, using generating-function equations [11]. This compares two given probabilistic systems; it is not a finite-realization existence theorem.

The sharper contrast is nonprobabilistic semantic finiteness. Jančar proves that, given a pushdown configuration, deciding whether it is bisimilar to some *unspecified* finite-state process is decidable [12]; Göller and Parys later show an elementary bound and decision procedure [13]. Our reusable local-pattern problem is realization-flavored and restricted to rule-native coordinates, while the seed problem is probabilistic and allows arbitrary hidden CTMCs, so neither theorem is a strict generalization of pushdown bisimulation finiteness. Nevertheless the decidable pushdown result shows that undecidability in [theorems 6.2](#) and [6.5](#) was not a foregone consequence of an unbounded stack-like state space. The crucial additional ingredient is the Kappa free-site zero test used in the two-counter encoding. We did not find in this literature the rule-language-specific statement of [theorem 2.2](#), nor the two exact-realization decision problems formulated here; those remain the claims for which priority requires specialist verification.

Kappa exact reductions. Internal coarse-graining, abstract differential semantics, intrinsic information carriers, stochastic fragments, and later static analyses already ask which correlations must be retained [15, 7, 18, 16]. The Kappa platform itself emphasizes local site-graph rules and network-free stochastic analysis [3]. Our negative theorems complement these constructive reductions: some finite local programs require an infinite predictive state space from one seed, and some reusable exact states cannot be expressed by any finite local pattern vocabulary.

Queueing and stochastic Petri nets. The reversible-tip process is not a new infinite-state Markov phenomenon. M/M/1 busy periods are classical [27], and stochastic Petri nets have long been used as finite rule-like specifications of Markov processes and queueing/performance models [24, 25]. The point of [section 3](#) is therefore only to exhibit the obstruction within ordinary biochemical site-graph syntax and to connect its non-rational assay law to exact-reduction assumptions.

Rule algebra. Rule-algebraic stochastic mechanics explains the finite overlap identities for graph-pattern observables used in [theorem 2.1](#) [1, 2]. We do not claim that overlap algebra as new.

ODE analogues. Observation-constrained maximal reduction is also established in deterministic kinetics. CLUE, for example, computes smallest-dimensional constrained linear lumpings preserving user-specified observables [23]. This is another reason not to market observation-seeded lumpability itself as the contribution.

8 Kappa/BNGL construct correspondence

The negative witnesses use no exotic language extension. Their ingredients correspond directly to standard rule-based constructs:

Mathematical ingredient	Kappa representation	BNGL/NFsim representation
Finite typed agent	agent with finite site signature	molecule type with finite components
Internal marker (tip, live, done)	site internal state	component internal state
Polymer adjacency	explicit site-site link	explicit bond labels between components
Free-site zero test	root site required free	unbound component in reactant pattern
Push / synthesis	agent present only on RHS	molecule present only in product pattern
Local count assay	Kappa pattern observable	BNGL <code>Molecules</code> observable
Network-free execution	KaSim	NFsim

The proofs are stated in Kappa-style site-graph semantics because that literature provides the cleanest formal rewriting framework. Whenever only the constructs in this table are used, the induced CTMC argument transfers directly to the corresponding BNGL/NFsim model; software-specific normalization is absorbed into the exact match hazard $a_r(x, m)$ in [equation \(12\)](#).

9 Implications for signaling and single-cell genomics

The two witnesses isolate different failure modes that matter in mechanistic biology.

The processive scanner is a processive-enzyme picture: a ribosome, polymerase, helicase, motor, or processive modification complex can move along a homogeneous template while the assay reports only a local terminal event. Exact prediction may require a nonlocal position/distance variable that no fixed local pattern vocabulary can reconstruct. A single integer counter is enough, but it is not a finite local site-graph statistic.

The reversible-tip example is a reversible assembly/disassembly picture: a polymer or oligomer tip can fluctuate by local addition and removal. Probabilistically it is the classical M/M/1 busy period, not a new queueing model. Its utility here is structural rather than falsification-based. Even when the exact mechanistic coordinate is just one integer (active length), the completion-time density has the half-integer asymptotic in [theorem 3.5](#); no finite-state kinetic scheme can reproduce that law exactly. Phase-type distributions are nevertheless dense, so finite noisy data on a bounded window cannot falsify finite-state kinetics from this tail alone. The exact statement is instead an approximation-order effect: along any sequence that converges to the true law, phase order must become unbounded. Thus increasing fitted order as the demanded approximation accuracy and observation horizon grow is consistent with structural infinite-state dynamics, but is not by itself a finite-data proof of them.

For signaling, analogous hidden coordinates include receptor-cluster size, processive phosphorylation position, polymerized scaffold length, or ordered modification context. For single-cell genomics, an RNA/ATAC observation operator may be finite-dimensional while the mechanistic predictive state beneath it is not. The rule-level closure test is [theorem C.2](#); finite assay dimension alone says nothing about finite mechanistic realizability.

This distinction is useful for NFsim/BNGL modeling. Network-free simulation can handle unbounded graph structure without enumerating all species, but a downstream inference pipeline should not assume that a finite list of BNGL observables is an exact state merely because those observables are biologically interpretable. When exact closure fails, one must either add a nonlocal counter/latent state, accept an infinite-state representation, or use approximation.

10 Conclusions

The main boundary is not “finite rules versus infinite rules.” It is local syntax versus exact predictive information.

1. The processive scanner gives a concrete finite Kappa/BNGL-style program for which no finite family of ordinary finite local patterns determines the future assay law over all reachable restarts. One nonlocal integer—distance remaining—is sufficient, so finite coordinate dimension and finite local expressibility are different notions.
2. The scanner’s designated seed nevertheless has a three-state exact realization. Reusable reduction and seed-specific realization are therefore genuinely different questions, not interchangeable quantifiers.
3. A reversible-tip site-graph program realizes the classical M/M/1 busy period. Its square-root transform rules out every finite-dimensional time-homogeneous linear realization from one finite seed. The associated $t^{-3/2}e^{-\eta t}$ tail is an exact asymptotic manifestation of that obstruction; phase-type denseness prevents finite-data falsification, but any convergent phase-type approximation sequence must use unbounded order.
4. Bounded complex size guarantees finite-dimensional exact species-count coordinates, while unbounded complexes may still admit simple integer coordinates. Finite dimension, finite local syntax, and finite state cardinality remain distinct axes.
5. Designated-seed finite hidden-CTMC realization and reusable finite-local-pattern assay sufficiency are undecidable for finite ordinary-Kappa programs. In both branches of the reductions the coarsest observation-compatible strong quotient is countably infinite, so the separation does not reduce to microscopic boundedness. This contrasts with decidability of bisimulation finiteness for ordinary pushdown processes.

For mechanistic inference, the practical consequence concerns exactness and model order, not finite-sample falsification. A finite list of biologically interpretable local observables need not be an exact state, and a finite-state hidden kinetic model need not exist even when every microscopic rule is local and elementary. Finite-state phase-type models can approximate such laws arbitrarily well, but no bounded phase order can converge to the reversible-tip law. Thus persistent pressure toward increasing fitted state count as accuracy demands tighten is compatible with a structural infinite-state mechanism rather than merely a poorly tuned finite model. Exact modeling then requires nonlocal state, an infinite-state representation, or an explicit approximation commitment.

A Rule-based stochastic semantics

Fix a finite Kappa/BNGL signature: finitely many agent/molecule types, finitely many sites per type, finitely many internal states per site, and finitely many site-graph rewrite rules. Let x_0 be a

finite seed and

$$\mathbf{S} := \text{Reach}(x_0)$$

the reachable mixtures modulo the graph-isomorphism convention of the stochastic semantics. Because every reachable mixture is finite and the signature is finite, $\mathbf{S}$ is finite or countably infinite.

For a rule r , let $\text{Match}_r(x)$ be its legal matches in mixture x , $T_r(x, m)$ the post-rewrite mixture, and $a_r(x, m) \geq 0$ the exact stochastic hazard assigned to match m . The generator is

$$(\mathcal{L}f)(x) = \sum_r \sum_{m \in \text{Match}_r(x)} a_r(x, m) [f(T_r(x, m)) - f(x)]. \quad (12)$$

This absorbs matching, automorphism, symmetry, and local-rate conventions into a_r rather than imposing a new convention. NFsim and KaSim are network-free simulators of such rule-induced stochastic semantics [29, 3].

Write $q(x, y)$ for the induced Q -matrix and

$$q_x = \sum_{y \neq x} q(x, y).$$

Unless stated otherwise, Q is stable ($q_x < \infty$ for every x), conservative, and nonexplosive.

B Observation-seeded strong lumping on a countable CTMC

Let $\mathcal{P}_0$ be the partition induced by the assay h . For a block D and $x \notin D$, define the finite off-diagonal rate measure

$$\nu_x(D) = \sum_{y \in D} q(x, y).$$

For a partition $\mathcal{P}$, refine each block by the vector of rates into all other $\mathcal{P}$ -blocks. Write

$$\mathcal{P}_{n+1} = \mathfrak{R}(\mathcal{P}_n), \quad x \sim_\omega x' \iff x \sim_n x' \text{ for all } n < \omega,$$

and let $\mathcal{P}_\omega$ be the intersection partition.

Proposition B.1 (The ω limit is enough for stable countable CTMCs). *Let Q be a stable conservative Q -matrix on a countable reachable state space. Then $\mathcal{P}_\omega$ is the coarsest strong lumping refining $\mathcal{P}_0$. No ordinal stage beyond ω is needed.*

Proof. Take x, x' in the same $\mathcal{P}_\omega$ block C and a different $\mathcal{P}_\omega$ block D . Choose $z \in D$ and let $D_n = [z]_{\mathcal{P}_n}$. Then $D_n \downarrow D$. Since $z \not\sim_\omega x$, there is N such that for $n \geq N$, D_n differs from the $\mathcal{P}_n$ block containing x, x' . Because $x \sim_{n+1} x'$,

$$\nu_x(D_n) = \nu_{x'}(D_n), \quad n \geq N.$$

The measures $\nu_x, \nu_{x'}$ are finite, so continuity from above gives

$$\nu_x(D) = \lim_n \nu_x(D_n) = \lim_n \nu_{x'}(D_n) = \nu_{x'}(D).$$

Hence $\mathcal{P}_\omega$ is strongly lumpable, and therefore $\mathfrak{R}(\mathcal{P}_\omega) = \mathcal{P}_\omega$ directly from the definition of the refinement operator.

For coarseness, let $\mathcal{R}$ be any strong lumping refining $\mathcal{P}_0$. Inductively suppose $\mathcal{R}$ refines $\mathcal{P}_n$. A $\mathcal{P}_n$ block is a countable union of $\mathcal{R}$ blocks. Strong lumpability gives equality of ν_x and $\nu_{x'}$ on every constituent $\mathcal{R}$ block for x, x' in one $\mathcal{R}$ block; countable additivity gives equality on the entire $\mathcal{P}_n$ block. Thus $\mathfrak{R}$ cannot split an $\mathcal{R}$ block, so $\mathcal{R}$ refines $\mathcal{P}_{n+1}$. Therefore $\mathcal{R}$ refines every $\mathcal{P}_n$ and hence $\mathcal{P}_\omega$. $\square$

Remark B.2 (Sharpness on the processive scanner). *For the scanner sets B_r defined in [section 2](#), the observation-seeded refinement is strict at every finite stage. Restricted to $\bigcup_{r \geq 0} B_r$, after n refinement steps its classes are*

$$B_0, B_1, \dots, B_n, \bigcup_{r \geq n+1} B_r.$$

Indeed, at stage 0 the assay separates Done (B_0) from all B_r , $r \geq 1$. If the displayed form holds at stage n , every state in B_{n+1} has a rate- κ transition into B_n , whereas every state in B_r , $r \geq n+2$, transitions into the tail class. Thus B_{n+1} splits from the tail at stage $n+1$, while previously isolated classes remain isolated. Induction gives the displayed partition at every finite depth. Hence no finite stage is stable; the ω intersection separates all remaining-distance classes, and the proposition above guarantees stability there. This is a sharpness witness for the background ω -closure result, not a separate contribution.

Remark B.3 (Finite total rate, not finite branching). *A state may have infinitely many successors, for example with rates proportional to 2^{-n} . What closes the argument is finite-measure continuity from above. This contrasts with general infinitely branching non-probabilistic transition systems, where finite-depth bisimulation approximants can require larger ordinals. Related event-bisimulation and approximation constructions are classical in labelled-Markov-process theory [\[8, 5\]](#).*

C The exact Kappa/BNGL macro-event criterion

Let $\Phi : \mathcal{S} \rightarrow \mathcal{Z}$ be a proposed mechanistic statistic. For $z' \neq \Phi(x)$ define

$$\Lambda_\Phi(x, z') = \sum_r \sum_{m \in \text{Match}_r(x)} a_r(x, m) \mathbf{1}\{\Phi(T_r(x, m)) = z'\}. \quad (13)$$

This is purely rule-semantic: rule matches, applications, and their exact hazards, grouped by the resulting macrostate.

Lemma C.1 (Small-time aggregate expansion). *For a stable CTMC and any set D not containing x ,*

$$\mathbb{P}_x\{X_t \in D\} = t\nu_x(D) + o(t) \quad (t \downarrow 0),$$

even when D is infinite.

Proof. Let Y_1 be the first-jump target. Conditional on a first jump to y by time t , the probability of at least one further jump before t is at most $1 - e^{-q_y t}$. Therefore

$$\mathbb{P}_x\{N_t \geq 2\} \leq q_x t \mathbb{E}[1 - e^{-q_{Y_1} t}] = o(t),$$

by dominated convergence under the first-jump target distribution. For exactly one jump into D , integrate the first-jump density and the no-second-jump factor $e^{-q_y(t-s)}$ over $y \in D$; dominated convergence over the same target distribution gives $t\nu_x(D) + o(t)$. Adding the $o(t)$ probability of two or more jumps proves the claim. $\square$

Proposition C.2 (Macro-event criterion). *The process $\Phi(X_t)$ is a time-homogeneous CTMC for every deterministic microscopic initial state, with one generator depending only on the value $\Phi(x)$, if and only if*

$$\Phi(x) = \Phi(x') \implies \Lambda_\Phi(x, z') = \Lambda_\Phi(x', z') \quad \forall z' \neq \Phi(x). \quad (14)$$

When this holds, $\bar{q}(z, z') = \Lambda_\Phi(x, z')$ for any representative x with $\Phi(x) = z$.

Proof. Sufficiency is strong lumpability of the fibers $\Phi^{-1}(z)$ because [equation \(13\)](#) is exactly the total microscopic transition rate into fiber $\Phi^{-1}(z')$. Conversely, if $Z_t = \Phi(X_t)$ is a CTMC with generator $\bar{q}$ for every deterministic initial state, then for $z = \Phi(x)$ and $z' \neq z$, [theorem C.1](#) gives

$$\mathbb{P}_x\{Z_t = z'\} = t\Lambda_\Phi(x, z') + o(t),$$

while the macro-CTMC expansion gives $t\bar{q}(z, z') + o(t)$. Hence $\Lambda_\Phi(x, z') = \bar{q}(z, z')$, independent of the representative. $\square$

Remark C.3 (BNGL rate laws). *Shared parameters, reversible rule pairs, BNGL local functions, and nonlinear functional rate laws do not affect [theorem C.2](#); all details are already inside $a_r(x, m)$. Rulewise decompositions may be convenient in special parameterizations but are not required.*

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